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1  -- =====
2  -- SECTION C : Aggregation Queries
3  -- Database   : Celebal_Week2
4  -- Table      : dbo.Superstore
5  -- Tool       : SQL Server Management Studio (SSMS)
6  -- Author     : ARATI THORAT | Celebal Internship - Week 2
7  -- =====
8
9  USE Celebal_Week2;
10 GO
11
12 -- =====
13 -- C1. Overall Business Summary
14 -- PURPOSE : Get a single-row KPI snapshot of the entire store
15 -- =====
16 SELECT
17     COUNT(*)                AS Total_Transactions,
18     COUNT(DISTINCT Order_ID) AS Total_Orders,
19     COUNT(DISTINCT Customer_ID) AS Total_Customers,
20     COUNT(DISTINCT Product_ID) AS Total_Products,
21     ROUND(SUM(Sales), 2)     AS Total_Revenue,
22     ROUND(SUM(Profit), 2)    AS Total_Profit,
23     SUM(Quantity)           AS Total_Units_Sold,
24     ROUND(AVG(Sales), 2)     AS Avg_Sale_Per_Transaction,
25     ROUND(AVG(Profit), 2)    AS Avg_Profit_Per_Transaction,
26     ROUND(MIN(Sales), 2)     AS Min_Sale,
27     ROUND(MAX(Sales), 2)     AS Max_Sale
28 FROM dbo.Superstore;
29
30 /*
31  OBSERVATION:
32  Total Revenue  ≈ $2.30 Million
33  Total Profit   ≈ $286,000
34  Overall profit margin ≈ 12.5%
35  9,994 transactions across 5,009 unique orders
36  Average sale per transaction ≈ $230
37 */
38
39
40 -- =====
41 -- C2. Sales and Profit by Category
42 -- PURPOSE : Compare revenue and profitability by product line
43 -- =====
44 SELECT
45     Category,
46     COUNT(*)                AS Total_Transactions,
47     SUM(Quantity)           AS Units_Sold,
48     ROUND(SUM(Sales), 2)     AS Total_Sales,
49     ROUND(SUM(Profit), 2)    AS Total_Profit,

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50     ROUND(AVG(Sales), 2) AS Avg_Sale,
51     ROUND(AVG(Profit), 2) AS Avg_Profit,
52     ROUND(MAX(Sales), 2) AS Max_Sale,
53     ROUND(MIN(Sales), 2) AS Min_Sale
54 FROM   dbo.Superstore
55 GROUP BY Category
56 ORDER BY Total_Sales DESC;
57
58 /*
59     OBSERVATION:
60     1. Technology → Highest revenue AND highest profit
61     2. Furniture → High revenue but LOW profit (losses on Tables)
62     3. Office Supp. → Most transactions, moderate profit
63     KEY INSIGHT: Furniture's high revenue doesn't translate to profit
64     due to excessive discounting on Tables and Bookcases.
65 */
66
67
68 -- _____
69 -- C3. Sales and Profit by Sub-Category
70 --     PURPOSE : Identify best and worst performing sub-categories
71 -- _____
72 SELECT
73     Category,
74     Sub_Category,
75     COUNT(*) AS Transactions,
76     ROUND(SUM(Sales), 2) AS Total_Sales,
77     ROUND(SUM(Profit), 2) AS Total_Profit,
78     ROUND(AVG(Discount), 4) AS Avg_Discount,
79     ROUND(SUM(Profit) * 100.0
80         / NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct
81 FROM   dbo.Superstore
82 GROUP BY Category, Sub_Category
83 ORDER BY Total_Profit DESC;
84
85 /*
86     OBSERVATION:
87     TOP PROFIT: Copiers, Phones, Accessories, Paper
88     LOSS-MAKING: Tables (-$17,725), Bookcases (-$3,473), Supplies (-$1,189)
89     Tables have an average discount > 40% → direct cause of losses.
90     Copiers have the highest profit margin at ~37%.
91 */
92
93
94 -- _____
95 -- C4. Sales and Profit by Region
96 --     PURPOSE : Identify strongest and weakest geographic markets
97 -- _____
98 SELECT

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99     Region,
100     COUNT(DISTINCT Order_ID) AS Total_Orders,
101     COUNT(DISTINCT Customer_ID) AS Total_Customers,
102     ROUND(SUM(Sales), 2) AS Total_Sales,
103     ROUND(SUM(Profit), 2) AS Total_Profit,
104     ROUND(AVG(Sales), 2) AS Avg_Order_Sale,
105     ROUND(SUM(Profit) * 100.0
106           / NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct
107 FROM   dbo.Superstore
108 GROUP BY Region
109 ORDER BY Total_Sales DESC;
110
111 /*
112  OBSERVATION:
113  1. West    → Highest sales ($725K) and profit ($108K)
114  2. East    → Second in sales and profit
115  3. Central → Lowest profit despite moderate sales (margin only ~7%)
116  4. South   → Lowest sales volume
117  Central region underperforms on profit – likely over-discounting.
118 */
119
120
121 -- -----
122 -- C5. Sales and Profit by Customer Segment
123 -- PURPOSE : Understand which segment drives the business
124 -- -----
125 SELECT
126     Segment,
127     COUNT(DISTINCT Customer_ID) AS Total_Customers,
128     COUNT(DISTINCT Order_ID) AS Total_Orders,
129     ROUND(SUM(Sales), 2) AS Total_Sales,
130     ROUND(SUM(Profit), 2) AS Total_Profit,
131     ROUND(AVG(Sales), 2) AS Avg_Sale_Per_Transaction,
132     ROUND(SUM(Profit) * 100.0
133           / NULLIF(SUM(Sales),0), 2) AS Profit_Margin_Pct
134 FROM   dbo.Superstore
135 GROUP BY Segment
136 ORDER BY Total_Sales DESC;
137
138 /*
139  OBSERVATION:
140  Consumer → 52% of total revenue (largest segment)
141  Corporate → Higher average order value, better margins
142  Home Office → Fewest customers but decent profit margin
143  RECOMMENDATION: Focus retention efforts on Corporate segment
144  as they generate the best profit per order.
145 */
146
147

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149 -- C6. Monthly Sales Trend (use case: monthly trends)
150 --     PURPOSE : Identify seasonal patterns in sales
151 -- 
152 SELECT
153     YEAR(Order_Date)                AS Order_Year,
154     MONTH(Order_Date)               AS Order_Month,
155     DATENAME(MONTH, Order_Date)     AS Month_Name,
156     COUNT(DISTINCT Order_ID)        AS Total_Orders,
157     ROUND(SUM(Sales), 2)            AS Monthly_Sales,
158     ROUND(SUM(Profit), 2)           AS Monthly_Profit,
159     ROUND(AVG(Sales), 2)            AS Avg_Sale
160 FROM   dbo.Superstore
161 GROUP BY
162     YEAR(Order_Date),
163     MONTH(Order_Date),
164     DATENAME(MONTH, Order_Date)
165 ORDER BY Order_Year, Order_Month;
166 
167 /*
168     OBSERVATION:
169     Clear seasonality pattern:
170     → Q4 (Oct, Nov, Dec) is consistently the highest-sales period
171     → Q1 (Jan, Feb) is the slowest period each year
172     → November and December spike due to holiday purchasing
173     This is valuable for inventory and staffing planning.
174 */
175 
176 
177 -- 
178 -- C7. Yearly Sales Summary
179 --     PURPOSE : Track year-over-year business growth
180 -- 
181 SELECT
182     YEAR(Order_Date)                AS Order_Year,
183     COUNT(DISTINCT Order_ID)        AS Total_Orders,
184     COUNT(DISTINCT Customer_ID)     AS Total_Customers,
185     ROUND(SUM(Sales), 2)            AS Annual_Sales,
186     ROUND(SUM(Profit), 2)           AS Annual_Profit,
187     ROUND(AVG(Sales), 2)            AS Avg_Sale
188 FROM   dbo.Superstore
189 GROUP BY YEAR(Order_Date)
190 ORDER BY Order_Year;
191 
192 /*
193     OBSERVATION:
194     Sales grew consistently from 2014 to 2017.
195     2017 shows the highest revenue (~$733K in available months).
196     Customer base also grew year-over-year, confirming business expansion.

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200 -- -----
201 -- C8. Top 10 Best-Selling Products (by Revenue)
202 --     PURPOSE : Identify hero products that drive revenue
203 -- -----
204 SELECT TOP 10
205     Product_ID,
206     Product_Name,
207     Category,
208     Sub_Category,
209     COUNT(*)                AS Times_Ordered,
210     SUM(Quantity)           AS Total_Units_Sold,
211     ROUND(SUM(Sales), 2)     AS Total_Revenue,
212     ROUND(SUM(Profit), 2)    AS Total_Profit
213 FROM   dbo.Superstore
214 GROUP BY Product_ID, Product_Name, Category, Sub_Category
215 ORDER BY Total_Revenue DESC;
216
217 /*
218     OBSERVATION:
219     Top revenue products are Copiers and Phones (Technology).
220     Canon imageCLASS Copier alone generates significant revenue.
221     These high-ticket items drive disproportionate profit.
222 */
223
224
225 -- -----
226 -- C9. Top 10 Customers by Total Purchase Value
227 --     PURPOSE : Identify VIP customers for retention strategy
228 -- -----
229 SELECT TOP 10
230     Customer_ID,
231     Customer_Name,
232     Segment,
233     Region,
234     COUNT(DISTINCT Order_ID) AS Total_Orders,
235     SUM(Quantity)           AS Total_Units_Bought,
236     ROUND(SUM(Sales), 2)     AS Total_Spent,
237     ROUND(SUM(Profit), 2)    AS Total_Profit_Generated
238 FROM   dbo.Superstore
239 GROUP BY Customer_ID, Customer_Name, Segment, Region
240 ORDER BY Total_Spent DESC;
241
242 /*
243     OBSERVATION:
244     Top customers are mostly in Consumer and Corporate segments.
245     Some top-spending customers actually generate negative profit
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246     due to excessive discounts – needs review.
247 */
248
249
250 -- _____
251 -- C10. Ship Mode Analysis
252 --     PURPOSE : Understand shipping preferences and their timing
253 -- _____
254 SELECT
255     Ship_Mode,
256     COUNT(DISTINCT Order_ID) AS Total_Orders,
257     ROUND(SUM(Sales), 2) AS Total_Sales,
258     ROUND(AVG(Sales), 2) AS Avg_Sale,
259     AVG(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Avg_Ship_Days,
260     MIN(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Min_Ship_Days,
261     MAX(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Max_Ship_Days
262 FROM     dbo.Superstore
263 GROUP BY Ship_Mode
264 ORDER BY Total_Orders DESC;
265
266 /*
267     OBSERVATION:
268     Standard Class → 60%+ of all orders, ships in 4–5 days average
269     Second Class   → ~20% of orders, ships in 3–4 days
270     First Class    → ~15% of orders, ships in 2 days
271     Same Day       → <5% of orders, ships in 0 days
272     Premium shipping modes are used less often but carry higher-value
273     orders.
274 */
275
276 -- _____
277 -- C11. State-Level Performance (Top 15 States)
278 --     PURPOSE : Geographic performance deep-dive
279 -- _____
280 SELECT TOP 15
281     State,
282     Region,
283     COUNT(DISTINCT Order_ID) AS Total_Orders,
284     ROUND(SUM(Sales), 2) AS Total_Sales,
285     ROUND(SUM(Profit), 2) AS Total_Profit,
286     ROUND(SUM(Profit) * 100.0
287           / NULLIF(SUM(Sales), 0), 2) AS Profit_Margin_Pct
288 FROM     dbo.Superstore
289 GROUP BY State, Region
290 ORDER BY Total_Sales DESC;
291
292 /*
293     OBSERVATION:
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294     California → Highest sales state (~$457K)
295     New York   → Second highest
296     Texas      → High sales but NEGATIVE total profit (-$25K)
297     Ohio       → Also shows loss despite decent sales volume
298     Texas and Ohio are key states to investigate for margin improvement.
299 */
300
301
302 -- -----
303 -- C12. Average Discount by Sub-Category
304 --     PURPOSE : Identify where the company over-discounts
305 -- -----
306 SELECT
307     Sub_Category,
308     ROUND(AVG(Discount), 4) AS Avg_Discount,
309     ROUND(SUM(Sales), 2) AS Total_Sales,
310     ROUND(SUM(Profit), 2) AS Total_Profit,
311     COUNT(*) AS Transactions
312 FROM     dbo.Superstore
313 GROUP BY Sub_Category
314 ORDER BY Avg_Discount DESC;
315
316 /*
317     OBSERVATION:
318     Binders, Tables, and Bookcases have the highest average discounts.
319     All three are associated with losses or thin profit margins.
320     INSIGHT: There is a direct correlation between high discount rates
321     and negative profitability in this dataset.
322 */
```